

PART III

ENCAPSULATION, POLYMORPHISM, AND INHERITANCE

The three main tenets of object-oriented programming are encapsulation, polymorphism, and inheritance. The next three chapters will explain each of these in turn, describing the underlying concepts and showing examples of how they are implemented in Python. In order for a programming language to call itself an OOP language, it must support all three of these central requirements. (If you're ever asked in an interview about the requirements for a language to be object-oriented, here's an easy way to remember them: it's as easy as PIE!)

Chapter 8 explains encapsulation: hiding the details and keeping everything in one place.

Chapter 9 discusses polymorphism: how multiple classes can have methods with the same names.

Chapter 10 covers inheritance: building on code that already exists.

Finally, Chapter 11 goes into detail on a number of topics (mostly dealing with memory management) that do not logically fit into the previous three chapters but are useful and important to OOP.